NeuralIntent: Bridging Noninvasive EEG and Embodied Foundation Models with Continuous Neural Intent Tokens

Anonymous submission

Abstract

Individuals with severe motor or communication impairments may retain clear object goals but lack a reliable way to convey them to assistive robots. We present NeuralIntent, a framework that maps noninvasive electroencephalography (EEG) into continuous Neural Intent Tokens (NITs) and calibrates them to condition an embodied foundation model. NeuralIntent maps the EEG-derived representation into the policy condition space through an embodiment-specific adapter. The EEG-derived representation specifies the target object, while the embodied policy generates the actions needed to interact with it. We introduce the Embodied Intent Grounding Benchmark (EIG-Bench), a controlled benchmark for evaluating whether EEG-derived target representations can guide embodied policies in multi-object scenes. On THINGS-EEG2, Qwen3-VL Layer 6 supervision improves retrieval among 200 candidates from 25.27% to 40.58%. Under strict leave-one-subject-out evaluation, calibrated NITs achieve 48.13% target selection accuracy. In a separate modular evaluation, the system achieves 65.63% task success, with all correctly selected objects subsequently grasped and lifted.

Introduction

Brain–computer interfaces (BCIs) provide an alternative pathway for people with severe motor or communication impairments to interact with the environment. Systems based on electroencephalography (EEG) have supported communication, assistive control, and rehabilitation by bypassing damaged neuromuscular pathways (Chaudhary, Birbaumer, and Ramos-Murguialday 2016; Biasiucci et al. 2018). However, noninvasive EEG has limited signal fidelity and information throughput relative to the demands of sustained trajectory- or joint-level robot control (Wolpaw et al. 2002; Millán et al. 2010). As manipulation tasks become more complex, requiring the user to continuously generate motor commands places an increasing burden on both neural decoding and the user.

Shared autonomy offers a more practical division of labor. Rather than specifying every movement, the user communicates a goal or preference, while the robot handles perception, planning, and execution (Dragan and Srinivasa 2013; Javdani et al. 2018). Recent AI-assisted BCIs have demonstrated that noninvasive neural signals can be combined with autonomous assistance for robotic manipulation (Lee et al. 2025). This suggests a different role for neural interfaces: instead of replacing the robot controller, they may provide the high-level information needed to determine which goal the robot should pursue. In assistive manipulation, the essential question is often not how every joint should move, but which object the user wants the robot to interact with.

Progress in visual EEG decoding provides a basis for recovering such object-related information. THINGS-EEG2 contains neural responses elicited by a large and diverse set of natural object images (Gifford et al. 2022). Contrastive EEG–image alignment enables large-scale zero-shot retrieval (Song et al. 2024), while subsequent work improves neural–visual alignment through better supervision and representations adapted to the limited fidelity of EEG (Zhang et al. 2026; Wu et al. 2026). These results show that visually evoked EEG contains recoverable object-level information. Nevertheless, existing evaluation largely ends with classification, retrieval, or reconstruction. Retrieving the correct image from an offline gallery leaves embodied policy consumption untested.

At the same time, vision–language–action and generalist robot policies can map visual observations and high-level task conditions to continuous robot actions (Zitkovich et al. 2023; Kim et al. 2025; NVIDIA et al. 2025). Their goals, however, are typically supplied through language instructions or predefined task descriptions. Conventional BCIs typically decode EEG into discrete target categories and then translate the predicted category into a robot command. We explore a representation-level interface in which target semantics recovered from EEG are preserved as a continuous condition and directly consumed by an embodied policy. Under this formulation, the neural decoder communicates what the user intends to manipulate, while the embodied model determines how to execute the task from its observations.

We introduce NeuralIntent, a framework that maps visually evoked EEG into a 512-dimensional **Neural Intent Token (NIT)** and calibrates this raw NIT into a continuous policy condition. The EEG encoder is supervised by intermediate semantic representations from Qwen3-VL, producing a continuous EEG-derived representation aligned with object semantics in a shared 512-D space (Bai et al. 2025). A policy-facing bridge calibrates the raw NIT, and an embodiment-specific adapter transforms the resulting 512-D condition into four continuous carrier tokens that are inserted into the multimodal input sequence of a GR00T N1.7-DROID policy (NVIDIA et al. 2025; NVIDIA 2026). The language instruction is fixed and target-agnostic across trials, making the calibrated NIT the only target-varying policy input during EEG-conditioned inference. The neural representation specifies the target object, while the embodied policy generates the corresponding actions.

We evaluate NeuralIntent with the Embodied Intent Grounding Benchmark (EIG-Bench), a controlled multi-object protocol that varies the target condition while holding the scene, instruction, policy, initial state, and randomness fixed. It reports target selection and subsequent execution as separate endpoints, distinguishing neural target errors from downstream execution failures.

The present study uses offline EEG recorded from healthy participants during passive viewing to evaluate visually evoked target semantics.

Our contributions are threefold:

- We introduce a representation-level EEG-to-policy interface that links subject-aware multi-level semantic alignment with embodiment-specific policy adaptation through continuous NITs, while optimizing the two stages separately.
- We introduce EIG-Bench to evaluate neural target conditioning under controlled interventions that hold language, policy, physical initialization, and randomness fixed.
- We evaluate semantic recovery, cross-participant transfer, target selection, and manipulation as distinct stages, separating neural target errors from downstream execution errors.

Related Work

Visual EEG Representation Learning

Visual EEG decoding has moved from closed-set classification toward alignment with pretrained visual spaces such as CLIP (Radford et al. 2021). On THINGS-EEG2, NICE established large-scale contrastive retrieval, ATM extended EEG embeddings to guided image reconstruction, and NeuroBridge introduced cognitive priors with bidirectional semantic alignment (Gifford et al. 2022; Song et al. 2024; Li et al. 2024; Zhang et al. 2026). A complementary line adapts the supervisory space itself through neural-aware prompt tuning, capacity-matched teachers, or intermediate visual features (Wang et al. 2025; Wu et al. 2026; Du et al. 2026; Jo et al. 2026). These methods recover object-related information, but their endpoints remain classification, retrieval, or reconstruction.

Recent multimodal and perceptual-prior models further broaden this alignment view (Zhang et al. 2025; Liu et al. 2026a). Neither line evaluates whether an EEG-derived condition can be consumed by a frozen embodied policy.

Cross-participant transfer is further complicated by subject-specific physiology and shifts in EEG feature distributions (Saha and Baumert 2020). NeuralIntent addresses this variability through subject-aware multi-level semantic alignment and asks whether a representation recovered from an unseen participant remains usable by an embodied policy. Related BCI work similarly treats cross-session and cross-subject robustness as a representation-transfer problem rather than a single-classifier problem (Tian et al. 2026).

Continuous Conditioning of Embodied Policies

Learned prefixes and input-conditioned prompts allow frozen transformers to consume continuous virtual tokens rather than discrete text alone (Li and Liang 2021; Zhou et al. 2022). In robotics, VIMA conditions a generalist manipulation policy with multimodal prompts (Jiang et al. 2023), while RT-2, OpenVLA, and GR00T map multimodal task conditions to robot actions (Zitkovich et al. 2023; Kim et al. 2025; NVIDIA et al. 2025). These works establish mechanisms for high-level policy conditioning; NeuralIntent asks whether the varying condition can instead be derived continuously from EEG. Recent work in embodied learning also treats alignment between multimodal semantic conditions and physical actions as an explicit design target (Qi et al. 2026).

Assistive BCIs and Brain–Robot Shared Autonomy

Shared autonomy combines partial user input with autonomous intent inference and low-level control (Dragan and Srinivasa 2013; Javdani et al. 2018). Brain–robot interfaces have used visually evoked EEG for candidate-object selection (Kolkhorst et al. 2020); more recent systems decode visual and motor imagery into object and placement choices (Liu et al. 2026b), or combine neural decoding with AI-assisted manipulation (Lee et al. 2025). These systems establish task-level brain–robot control but retain an explicit decision between neural decoding and action generation. NeuralIntent instead tests whether an EEG-derived continuous representation can directly condition an embodied policy. AAAI studies of shared autonomy additionally analyze intervention timing, emphasizing that assistance should preserve both task performance and user-directed goals (Tan et al. 2022).

Reliable policy evaluation also requires reproducible schedules and targeted tests of behavioral sensitivity. SIMPLER studies paired simulated and real-world evaluations, while robot contrast sets apply small, specific perturbations to otherwise matched test instances (Li et al. 2025; Anwar, Gupta, and Thomason 2025). EIG-Bench adopts a complementary condition-level intervention: it fixes the scene, language, policy, initial state, and randomness while replacing only the neural policy condition.

Method

Overview and Problem Formulation

NeuralIntent separates neural target representation from embodiment-specific action generation. It maps visually evoked EEG to a raw NIT, calibrates the representation in policy space, and converts it into carrier tokens that condition the embodied policy. Figure 1 summarizes this three-stage interface and its controlled evaluation.

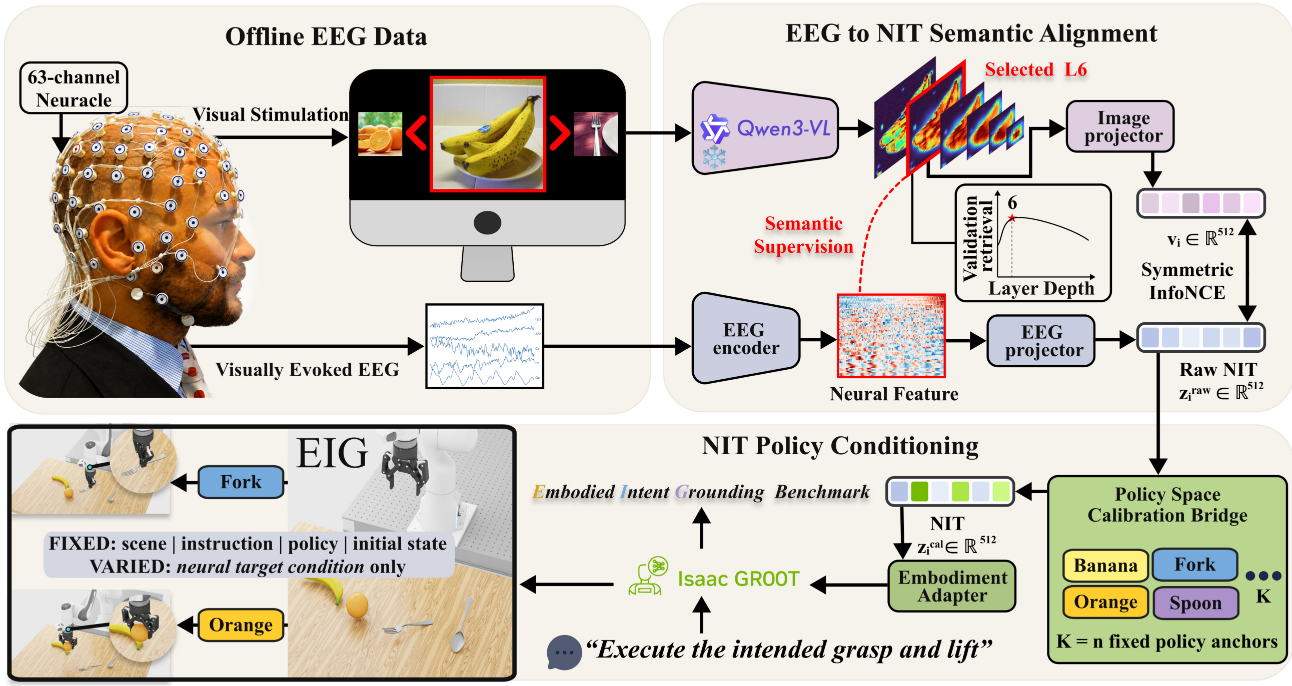

Figure 1: NeuralIntent overview. Offline passive viewing EEG is aligned with image semantics from a validation-selected Qwen3-VL layer to produce a 512-D raw NIT. A frozen policy-space calibration bridge maps this representation into the closed-set policy-condition space, after which an embodiment adapter injects the calibrated condition into GR00T together with the visual observation and neutral instruction. EIG-Bench holds the scene, instruction, policy, initial state, and randomness fixed while changing only the neural condition.

Formally, given an EEG segment x elicited by a viewed target, NeuralIntent produces a 512-D *raw NIT*, calibrates it to the policy condition space, and converts the result into embodiment-specific carrier tokens. Equation (1) summarizes the inference path for embodiment e:

$$\begin{aligned} z^{\text{raw}} &= F_{\text{EEG}}(x), \\ z^{\text{cal}} &= B(z^{\text{raw}}), \\ C_e &= G_e(z^{\text{cal}}) \in \mathbb{R}^{m \times d_e}, \\ a^{(e)}_{t:t+H} &\sim \pi_e(\cdot \mid o^{(e)}_t, l_0, C_e), \end{aligned} \tag{1}$$

where $o^{(e)}_t$ is the current visual and proprioceptive observation, l_0 is a fixed target-agnostic instruction, and π_e predicts an action chunk of horizon H. We instantiate $m = 4$ and $d_e = 2048$ with GR00T N1.7-DROID. The EEG encoder, calibration bridge, and policy adapter are optimized separately and frozen before formal rollout.

EEG-to-NIT Semantic Alignment

Single-layer semantic alignment. The first stage places EEG and its paired image in a shared semantic space. For image I, frozen Qwen3-VL-8B provides the supervision target (Bai et al. 2025). We mean-pool its image-token states at decoder layer ℓ, subtract the optimization-set mean, and apply ℓ_2 normalization to obtain $t_\ell(I)$. Controlled comparisons select $\ell = 6$ on the stimulus-disjoint validation split. A NICE-style spatiotemporal encoder (Song et al. 2024) and two projectors produce the paired embeddings in Equation (2):

$$\begin{aligned} z_i^{\text{raw}} &= \text{norm}(g_\phi(f_\psi(x_i; \Theta_{u_i}))) \,, \\ v_i &= \text{norm}(p_\theta(t_6(I_i))) \,, \end{aligned} \tag{2}$$

where Θ_u is the participant-specific spatial adapter. For B paired samples, we optimize symmetric InfoNCE (van den Oord, Li, and Vinyals 2018; Radford et al. 2021): the batch similarity matrix $S_{ij} = (z_i^{\text{raw}})^\top v_j / \tau$ is trained in both EEG-to-image and image-to-EEG directions. Qwen3-VL is frozen; only f_ψ, g_ϕ, p_θ, and the training-participant adapters are optimized. The output is a raw NIT in the learned semantic space, before policy-facing calibration.

We preselect the Qwen3-VL-8B supervision depth using the stimulus-disjoint validation split. The selected layer is fixed before any test or EIG-Bench evaluation. Results report the complete screen over six predefined candidate layers.

Subject-aware multi-level alignment. A single layer supports controlled teacher comparisons. For strict leave-one-subject-out (LOSO) transfer in the embodied evaluation, NeuralIntent instead learns participant-aware routing over $\mathcal{L} = \{6, 10, 14, 18\}$. Equation (3) defines the routed representations for participant u:

$$
\begin{aligned}
w_{\ell,u} &= \mathrm{softmax}_\ell\big((a_\ell + b_{\ell,u})/\tau_r\big),\\
v_u(I) &= \mathrm{norm}\left(P_s \sum_{\ell\in\mathcal{L}} w_{\ell,u} P_\ell t_\ell(I)\right),\\
z^{\mathrm{raw}}_{u,\mathrm{ML}} &= \mathrm{norm}(P_s g_\phi(f_\psi(x;\Theta_u)))\,.
\end{aligned} \tag{3}
$$

Here P_ℓ projects each decoder layer to 512 dimensions, P_s is shared by both branches, and $\tau_r = 1$. Training uses participant and layer dropout. For fold $-u^*$, all parameters are learned without u^*. At inference, we set $b_{\ell,u^*} = 0$ and use the mean spatial adapter from the training participants. No held-out optimization is performed.

Policy-Facing Condition Construction and Calibration

Policy anchor bank and support bridge. The policy is adapted to visual semantic anchors, whereas raw NITs follow an EEG-dependent distribution. We therefore express EEG conditions in the same 512-D policy space before carrier injection. For each EIG-Bench concept c, a fixed support image I_c defines $q_c = \mathrm{norm}(p_\theta(t_6(I_c)))$, with $Q = [q_1, \dots, q_K]$ and $K = 4$. Let n_c be the raw NIT obtained by averaging the 80 EEG repetitions paired with I_c before encoding, and let $N = [n_1, \dots, n_K]$. A rank-four residual map is fitted from N to Q, and its normalized output defines the support condition p_c^{EEG}. Because N has full column rank, the fitted support columns reproduce Q to numerical precision. This condition therefore verifies the calibration and carrier path. The closed-form residual solution is given in the supplementary material.

Strict-LOSO calibration. Formal held-out conditions average the EEG waveform before nonlinear encoding. For each held-out participant–concept pair, we average the R repetitions in signal space and encode the mean with the corresponding strict-LOSO model. Let $V_{-u^*} = [v_{-u^*}(I_1), \dots, v_{-u^*}(I_K)]$ denote the visual anchors from the fold-specific multi-level branch. Equation (4) defines the complete calibration bridge, comprising residual alignment followed by anchor attention:

$$
\begin{aligned}
M_{-u^*} &= (Q - V_{-u^*})(V_{-u^*}^\top V_{-u^*})^\dagger V_{-u^*}^\top,\\
\widetilde{z}_{u^*,c} &= \mathrm{norm}\big[(I_{512} + M_{-u^*}) z^{\mathrm{raw}}_{u^*,c}\big]\,,\\
\alpha_k &= \mathrm{softmax}_k(\cos(\widetilde{z}_{u^*,c}, q_k)/\tau_a)\,,\\
z^{\mathrm{cal}}_{u^*,c} &= \mathrm{norm}\left(\sum_{k=1}^{K} \alpha_k q_k\right).
\end{aligned} \tag{4}
$$

The representation-source and modular campaigns use $\tau_a = 0.01$, whereas the independent three-router selection-only audit uses the validation-selected $\tau_a = 0.10$. Each value is fixed before its formal rollout and is never tuned on formal episodes. The calibration is closed-set over the four legal EIG-Bench targets. Policy optimization uses semantic anchors q_c exclusively.

Carrier-Conditioned Policy Adaptation

Carrier injection. To expose a 512-D neural condition to GR00T, the embodiment adapter uses Equation (5) to convert policy condition s_c into four 2048-D carrier tokens:

$$
\begin{aligned}
r_c &= W_2\,\mathrm{GELU}(W_1\,\mathrm{LN}(s_c))\,,\\
C_{\mathrm{arm}}(s_c) &= \gamma_{\max}\tanh(\rho)\ \mathrm{LN}\big(\mathrm{reshape}_{4\times 2048}(r_c)\big)\,.
\end{aligned} \tag{5}
$$

The bias-free projections use dimensions $512 \to 1024 \to 8192$; both layer normalizations are non-affine, and $\gamma_{\max} = 1$. Hence a zero condition has zero carrier content while retaining the same four sequence positions. Carriers are appended after the visual–language embeddings, assigned consecutive text-like M-RoPE positions, and included in causal attention. The instruction is fixed to "Execute the intended grasp and lift."

Execution and condition-router objectives. We separate execution competence from sensitivity to the injected condition. The inherited execution overlay retains GR00T's masked flow-matching loss ℓ_{FM} (Lipman et al. 2023; NVIDIA et al. 2025; NVIDIA 2026); its full definition is provided in the supplementary material. The condition-router stage pairs each correct anchor q_y with a wrong anchor q_{y^-} under identical diffusion noise and time. Equation (6) defines the resulting router objective:

$$
\mathcal{L}_\pi = \ell_{\mathrm{FM}}(q_y) + \lambda_{\mathrm{cf}}\left[\ell_{\mathrm{FM}}(q_y) - \ell_{\mathrm{FM}}(q_{y^-}) + \delta\right]_+\,, \tag{6}
$$

where $[r]_+ = \max(r, 0)$, $\lambda_{\mathrm{cf}} = 0.5$, and $\delta = 0.05$. This margin requires the correct condition to explain the demonstrated action better than a wrong condition. The execution overlay is frozen during this stage; only G_{arm} is updated. Formal rollouts freeze every module and vary only $s_c \in \{q_c, p_c^{\mathrm{EEG}}, z^{\mathrm{cal}}_{u,c}\}$ at initialization.

EIG-Bench

EIG-Bench tests whether substituting a continuous neural condition redirects object selection while all non-condition inputs remain fixed. Paired trials share the scene, target-agnostic instruction, policy checkpoint, random seed, and initial simulator state. Figure 1 summarizes this controlled intervention.

Balanced Paired Protocol

Banana, orange, fork, and spoon occupy a 0.56-m equal-reach front arc at $-27°$, $-9°$, $9°$, and $27°$. A cyclic Latin square rotates every object through all four positions, preventing position from encoding the target and fixing chance selection at 25%.

Two interventions test complementary questions. A *representation-source intervention* fixes the target while substituting the semantic anchor q_c, support condition p_c^{EEG}, or held-out condition $z^{\mathrm{cal}}_{u,c}$, testing whether each representation is consumable by the same policy. A *target-condition intervention* instead replays an exact initial state with another target condition or a code control, testing whether selection follows the injected condition rather than the unchanged scene.

Diagnostic Endpoints

Fixed-prefix target approach is assigned to the object nearest the end effector after a fixed policy prefix; wrong-object and no-selection outcomes are failures. A stricter radius–dwell endpoint requires the same nearest object to remain within 0.15 m for 15 consecutive frames. For modular execution, the fixed-prefix object is passed to a target-agnostic grasp-and-lift executor, where full success requires a stable 5-cm lift.

The policy, executor, and controller never receive the hidden evaluation target. We therefore report prefix selection, full-task success, and completion conditioned on correct selection. This diagnostic chain separates neural target errors from downstream manipulation failures across the shared-autonomy selection-to-lift chain.

Experimental Setup

EEG Dataset and Evaluation Protocol

We use the artifact-corrected THINGS-EEG2 release (Gifford et al. 2022): 10 participants, 63 channels, 250 Hz, and 250 time samples per epoch. Per participant, 16,140 of the 16,540 training stimuli are used for optimization and 400 stimulus-disjoint images for validation. The official test partition contains 200 concept-disjoint stimuli. After trial averaging, the pooled optimization, validation, and test sets contain 161,400, 4,000, and 2,000 samples. The 200-way test gallery has 0.50% Top-1 chance.

We report jointly trained seen-participant retrieval and strict LOSO transfer over all 10 held-out participants. Main anchor and fixed-capacity comparisons use three seeds, with validation-only checkpoint selection. Full split, checkpoint, single-trial, and prompt-bank null specifications appear in the supplementary material.

EIG-Bench Campaigns and Statistical Analysis

The three-router audit compares raw NIT, target-excluded auxiliary bridge, K4 soft calibration, and oracle-reference conditions. Each condition contains 160 episodes per router on a shared reserved schedule, yielding 1,920 episodes. We report fixed-prefix target approach and radius–dwell selection.

The representation-source campaign uses 10 unseen scene seeds, four Latin-square layouts, and four targets, yielding 160 episodes per condition and 480 total. Each held-out participant contributes 16 balanced episodes. Conditions sharing a pair key have identical non-condition inputs and matching initial-state hashes. This campaign uses its own frozen selector and reserved schedule.

The modular execution campaign uses reserved seeds and 160 episodes for each of two EEG-derived conditions, yielding 320 episodes. Its selector, continuation checkpoint, and target-agnostic inverse-kinematics controller are fixed before evaluation. Confidence intervals use 10,000 paired bootstrap samples, and binary outcomes use exact McNemar tests. Participant-clustered analyses treat each held-out participant as the neural statistical unit.

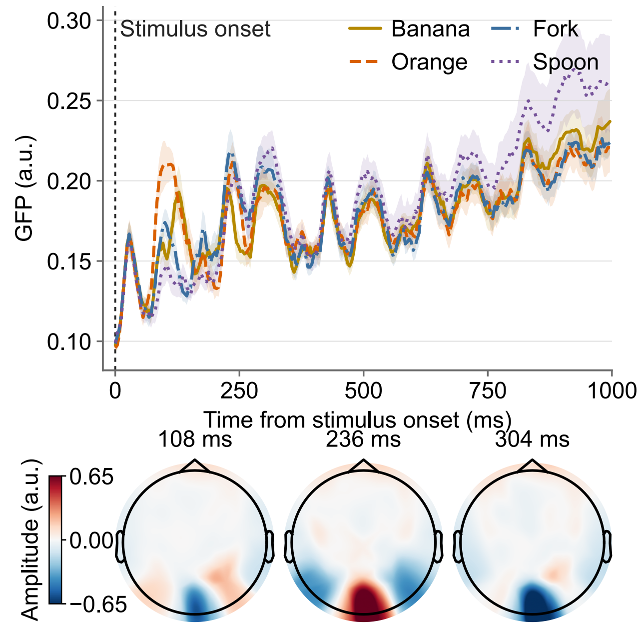

Figure 2: Target-conditional EEG responses. Curves show participant-mean GFP ± SEM ($n = 10$) after within-participant averaging; maps show grand-average scalp activity at three target-agnostic GFP peaks on a shared scale.

Decoder layer	L6	L10	L14	L18	L22	L26
Val. Top-1 (%)	**19.70**	18.83	17.95	18.00	17.05	17.75

Table 1: Predefined supervision depth screen. Validation Top-1 retrieval for the six predefined Qwen3-VL-8B candidate layers. L6 is selected before test evaluation because it attains the highest validation score.

Results

We organize the evaluation as a linked evidence chain: semantic recovery, cross-participant transfer, condition consumption by the policy, and downstream selection-to-lift execution. A distinct endpoint at each stage preserves stage-specific attribution.

Semantic Recovery and Cross-Participant Transfer

Before evaluating semantic recovery, we visualize the target-conditional EEG responses associated with the four EIG-Bench objects. Figure 2 shows event-locked temporal structure after within-participant trial averaging, while the strong overlap across targets illustrates the fine-grained nature of the decoding problem.

Semantic recovery. The Qwen3-VL-8B supervision depth is selected on the stimulus-disjoint validation split before test evaluation. Among the six predefined candidate layers, L6 gives the highest validation Top-1, so we use it for all subsequent single-layer comparisons and policy-anchor construction.

A. Seen-participant retrieval		
Anchor / target	Top-1 ↑	Top-5 ↑
Published per-participant protocols[†]		
NICE (CLIP) (Song et al. 2024)	16.10	43.60
ATM-S (CLIP) (Li et al. 2024)	27.10	58.10
NeuroBridge (CLIP) (Zhang et al. 2026)	63.20	89.90
Shallow Alignment (Du et al. 2026)	82.60	97.70
NeuralIntent anchors		
CLIP-B/32 (512-D)	25.27±0.56	56.62±0.22
Qwen3-VL L6	**40.58±0.42**	**74.97±1.65**
Qwen3-VL L6 (512-D)	33.45±0.94	67.87±0.90

B. Strict LOSO transfer		
Variant	Top-1 ↑	Top-5 ↑
Baselines		
CLIP baseline	11.95±2.43	34.35±5.28
Qwen-L6	15.80±2.69	41.80±4.22
Transfer variants		
Qwen-L6 + mean adapter	19.05±4.83	45.35±8.37
Multi-layer	16.25±2.87	41.25±5.71
Final NeuralIntent	**19.95±4.29**	**45.55±6.80**

Table 2: Semantic recovery on THINGS-EEG2 (%). (A) Seen-participant 200-way retrieval; (B) strict-LOSO transfer. NeuralIntent values are mean±SD over three seeds in (A) and across held-out participants in (B); published results are contextual because their protocols differ.

Table 2 summarizes the matched supervision-space and capacity controls, contextual published results, and strict-LOSO transfer.

Among the NeuralIntent anchors, Qwen-L6 supervision provides the strongest recovery in both the native and capacity-matched comparisons, retaining an 8.18-point advantage under matched projector capacity. The strict-LOSO block remains substantially harder than seen-participant retrieval, leaving a substantial cross-participant gap before the policy-facing stages. Complete anchor and seed-0 layer comparisons, together with encoder and multi-seed baseline controls, appear in the Supplementary Material; per-seed candidate-layer audits are retained in Source Data.

Strict cross-participant generalization. Figure 3 shows that the largest cumulative changes arise from the semantic teacher and mean adapter, whereas multi-level routing adds a smaller and more heterogeneous refinement. The participant spread and the 100% participant-ID probe confirm persistent participant-specific structure alongside cross-participant concept recovery. Complete participant-level Top-1/Top-5 results appear in Supplementary Table S3.

Construction ablations. Completed domain-alignment, repeat-consistency, and visual-prompt variants do not yield a consistent improvement over the selected construction. We retain the simplest validated configuration for subsequent experiments.

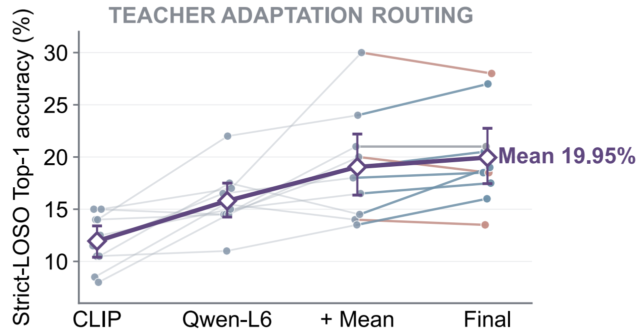

Figure 3: Strict-LOSO participant trajectories across NeuralIntent components. Lines denote held-out participants; diamonds show stage means.

Policy condition	Fixed-prefix approach ↑	Radius–dwell selection ↑
Raw strict-LOSO NIT	22.08±0.36	8.54±0.36
Auxiliary-196 bridge	38.13±0.63	11.46±0.36
K4 soft calibration ($\tau = 0.10$)	**58.75±0.00**	**24.79±0.95**
Oracle semantic reference	78.54±1.91	31.88±1.25

Table 3: Three-router condition-consumption audit (%). Mean±SD over three router seeds, with 160 paired episodes per condition and seed; chance is 25%. Both columns report selection-only endpoints.

Image and language addressability. The representation probes recover concept-related structure but also expose strong participant identity. Full-bank retrieval and the stricter target-held-out H4 test remain below the prespecified threshold. Policy evaluation therefore uses the closed-set NIT conditions defined above. Completed null analyses and coverage counts appear in the Supplementary Material.

Neural-Conditioned Target Approach and Selection

We first test whether condition consumption persists across independently trained condition routers. Table 3 compares four conditions under three staged routers and a fresh, shared selection-only schedule. The Auxiliary-196 bridge is fitted on non-EIG concepts and excludes the four EIG-Bench targets; the staged routers themselves are trained on the same four semantic-anchor demonstrations.

Table 3 shows that policy-facing calibration improves the fixed-prefix approach endpoint over raw and target-excluded conditions, while remaining below the oracle semantic reference. The stricter dwell endpoint stays weaker across conditions, showing that stable target commitment remains harder than fixed-prefix approach. Seed- and target-resolved breakdowns are provided in the Supplementary Material.

We next use a separate frozen-policy campaign to intervene only on the input condition under byte-identical initial physical states.

When only the injected NIT was cyclically permuted, 75.00% of paired episodes changed their selected object.

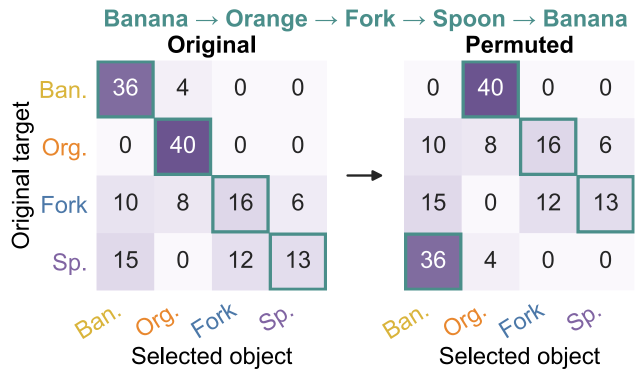

Figure 4: Exact-state condition permutation. Selection matrices before and after cyclically permuting only the NIT condition. Teal outlines mark condition-specified targets across 160 paired states.

Selection matched the new condition in 65.63% of episodes, while agreement with the original target fell to 12.50%. Since the scene, instruction, policy, initial state, and random seed were identical within each pair, the change in behavior can be attributed to the injected neural condition (Figure 4).

Under the same frozen policy and evaluation schedule, hard argmax and soft calibrated NIT conditions produced comparable agreement with the injected target (64.38% and 65.00% at $\tau_a = 0.01$, respectively), while increasing the temperature to $\tau_a = 0.10$ yielded 61.88%. The near-tie shows that soft and discretized conditions transfer the injected target information similarly in the current four-target setting. Shuffle and null controls appear in the Supplementary Material.

An independent radius–dwell campaign evaluates representation-source transfer. The held-out calibrated condition reaches 77/160 selections (48.13%) and exceeds chance in a participant-level sign-flip test ($p = 0.00195$); its gap to fitted support localizes the main loss to representation transfer and calibration. Target-resolved errors appear in the Supplementary Material.

Target Selection and Modular Execution

NeuralIntent separates EEG-conditioned target selection from target-agnostic execution. All correctly selected strict-LOSO NIT rollouts completed the grasp and lift sequence (105/105; Table 4). Figure 5 shows representative executions, and Supplementary Figure S3 provides a dual-camera audit of failure cases using recorded simulator states.

Limitations

NeuralIntent is evaluated using offline passive-viewing EEG from ten healthy participants and therefore does not yet demonstrate spontaneous intention decoding, online closed-loop control, or patient benefit. The current study evaluates neural policy conditioning in a controlled four-target setting. Evaluation on larger and unseen target sets remains future work. EIG-Bench evaluates a single simulated robot policy with a modular inverse-kinematics executor rather than end-to-end learned manipulation. Future work should validate target users, online operation, and transfer across policies and embodiments.

Condition	Prefix selection	Full task	Full task given correct
Support condition	141/160 (88.13)	141/160 (88.13)	141/141 (100.00)
Strict-LOSO NIT	**105/160 (65.63)**	**105/160 (65.63)**	**105/105 (100.00)**

Table 4: Core modular execution over 160 episodes per condition, reported as successes/episodes (%); the frozen executor receives only the prefix-selected object's live pose.

Conclusion

We presented NeuralIntent, a closed-set EEG-to-policy framework that maps visually evoked EEG into continuous Neural Intent Tokens (NITs) and calibrates them to condition an embodied policy without emitting an intermediate class label or text command. Across three independently trained condition routers, K4 calibration increased fixed-prefix target approach from 22.08% for raw strict-LOSO NITs to 58.75%. In exact-state interventions, changing only the NIT condition redirected object selection: selection followed the injected condition in 65.63% of episodes, agreement with the original target fell to 12.50%, and 75.00% of paired episodes switched objects. In the independent modular evaluation, strict-LOSO NITs selected the target object in 105 of 160 trials, after which the fixed executor completed all 105 grasp-and-lift tasks. These results show that EEG-derived conditions can serve as goal inputs to an embodied policy held fixed during evaluation. Cross-participant transfer and sustained target commitment remain the principal limitations.

Figure 5: Representative held-out executions. (a) A complete banana rollout. (b) Verified terminal lifts for the remaining targets. (c) Representative failures from the separate exact-state campaign.

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